

Project Description – Bright Coffee

Background

Bright Coffee Shop is a multi-location coffee retailer operating across three New York neighbourhoods: Astoria, Hell's Kitchen, and Lower Manhattan. Following the appointment of a new CEO focused on revenue growth and product performance, the business needed a data-driven view of how it was performing across stores, products, and time periods.

Role & Objective

Acting as a Junior Data Analyst, the goal of this project was to extract actionable insights from six months of historical sales data (January–May 2023) and translate them into a clear presentation that could support the CEO's decision-making. Specifically, the analysis set out to answer:

- Which products generate the most revenue
- What time of day the store performs best
- How sales trends move across products and time intervals
- What actions could improve overall sales performance

Data

The underlying dataset was sourced from the company's internal CRM (Sales Department) and contained roughly 149,000 transaction records across 20 columns, including transaction date, store location, transaction quantity, unit price, and product category/detail.

Methodology

The project followed a structured analytics workflow:

1. **Planning & Architecture** — Mapped out the data flow from source through processing, storage, and reporting, and defined the key insights and calculations the analysis needed to deliver.
2. **Data Processing** — Cleaned and transformed the raw data: corrected malformed unit price values (e.g., commas used instead of decimal points), converted

timestamps into usable formats, and feature engineered new columns such as time-of-day buckets, spend buckets, and a calculated revenue-per-row field (unit price × transaction quantity).

3. **Data Analysis** — Aggregated the cleaned data to revenue and unit-sold totals by product category, store location, time of day, and day of week.

4. **Presentation** — Combined the findings into a presentation covering the project overview, data preparation, key insights, and recommendations.

Key Findings

- Total revenue across the six-month period was **\$698,812.33**, from **214,470** units sold at an average of **\$4.68** per transaction.
- **Barista Espresso** was the best-selling individual product, and **coffee** was the top-performing category overall, followed by tea and bakery items while packaged chocolate sold the least.
- Revenue was nearly evenly split across the three store locations (each contributing roughly a third), with **Hell's Kitchen** narrowly leading. This suggests untapped growth potential in the other, comparably sized markets.
- Revenue grew steadily month over month, peaking in **June**, hinting at a seasonal change.
- **Mornings** drove the most revenue at nearly every location except Astoria (which performed well in the afternoon), while **evenings** consistently generated the least.
- **Weekdays** accounted for 72% of units sold versus 28% on weekends.

Recommendations

- Launch targeted marketing campaigns during slower sales periods (e.g., evenings and weekends) to lift traffic.
- Expand the menu with items suited to all times of day to generate more revenue across the full business day.
- Introduce seasonal promotions ahead of winter months to sustain the observed revenue growth trend.

- Assess the market by introducing higher-priced or premium product offerings in wealthier neighbourhoods where spending capacity appears underutilized.

Tools & Deliverables

The project was completed using Databricks for data preparation, Microsoft Excel for pivot tables, and charting, and PowerPoint presentation for stakeholder review.